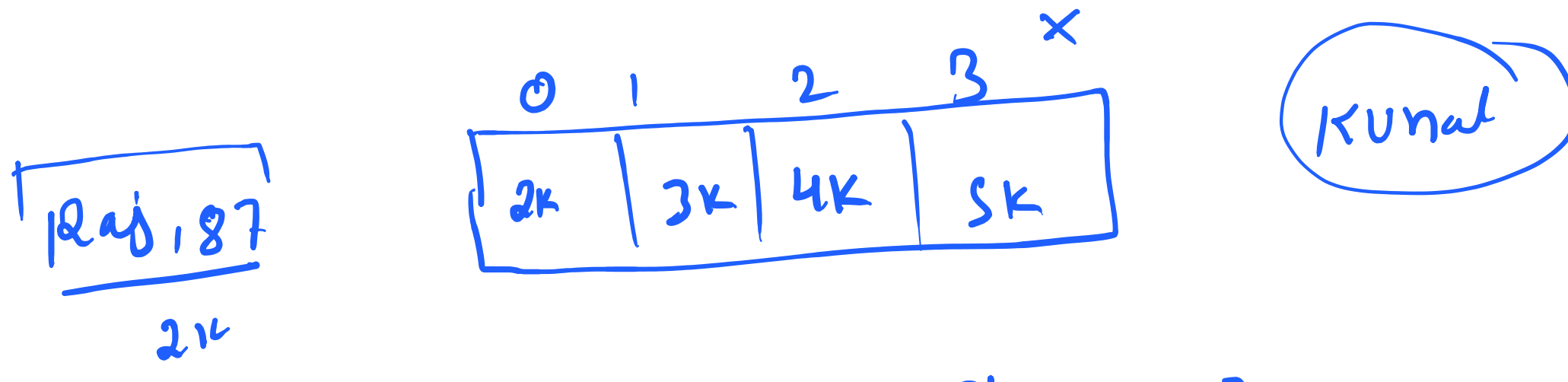


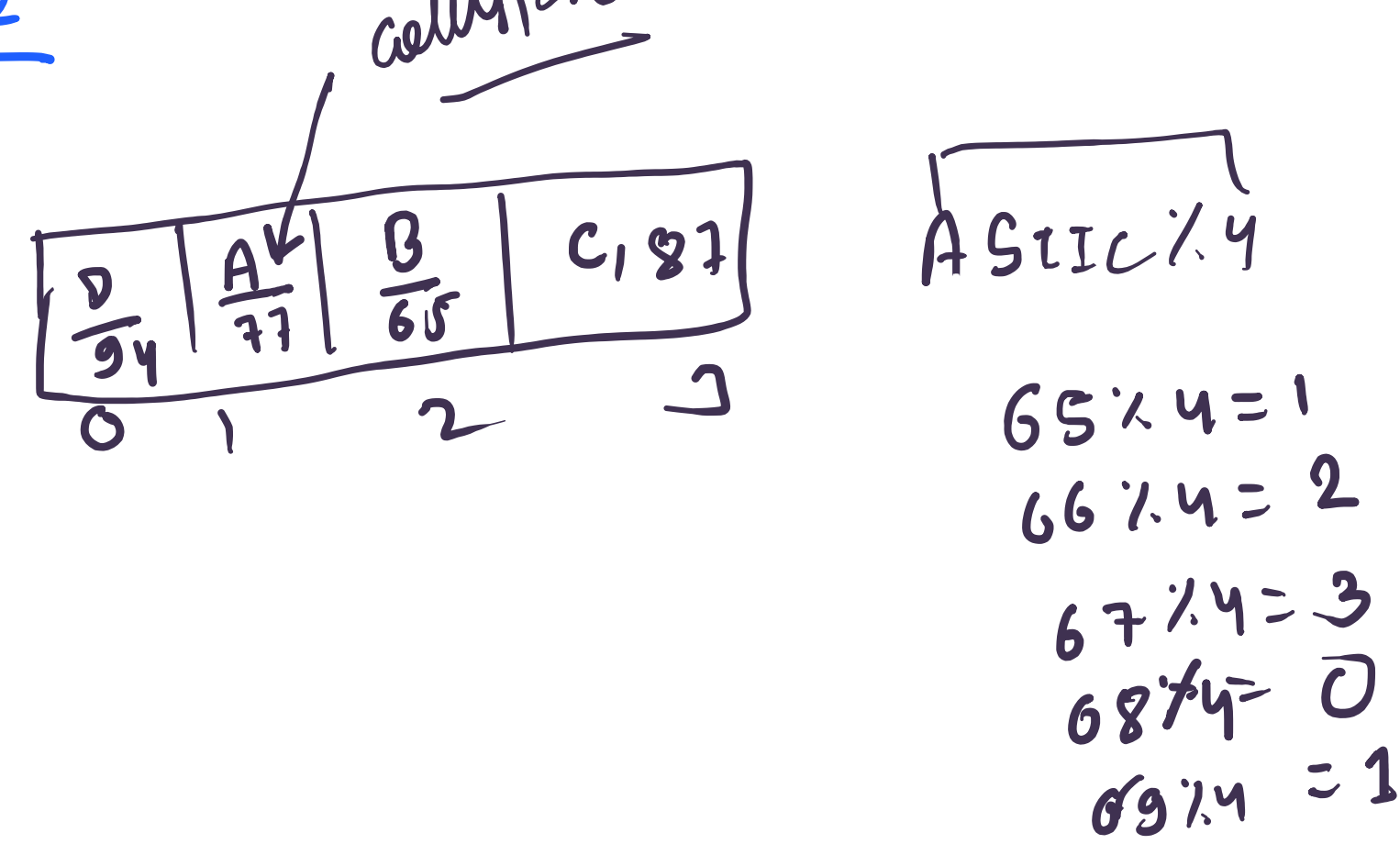
Hash Map

Key value pair $\rightarrow$ OCIO



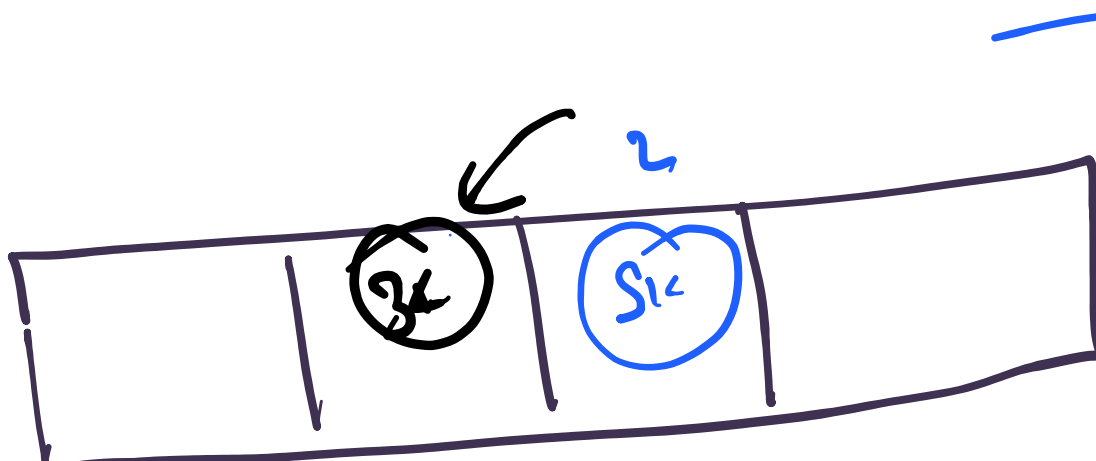
① $\text{hash fun} \rightarrow \text{object} \rightarrow \text{key} \rightarrow \text{unique value}$

A → 77
B → 65
C → 87
D → 94
E, 97



⑪ Chainers $\rightarrow$ LinkedList

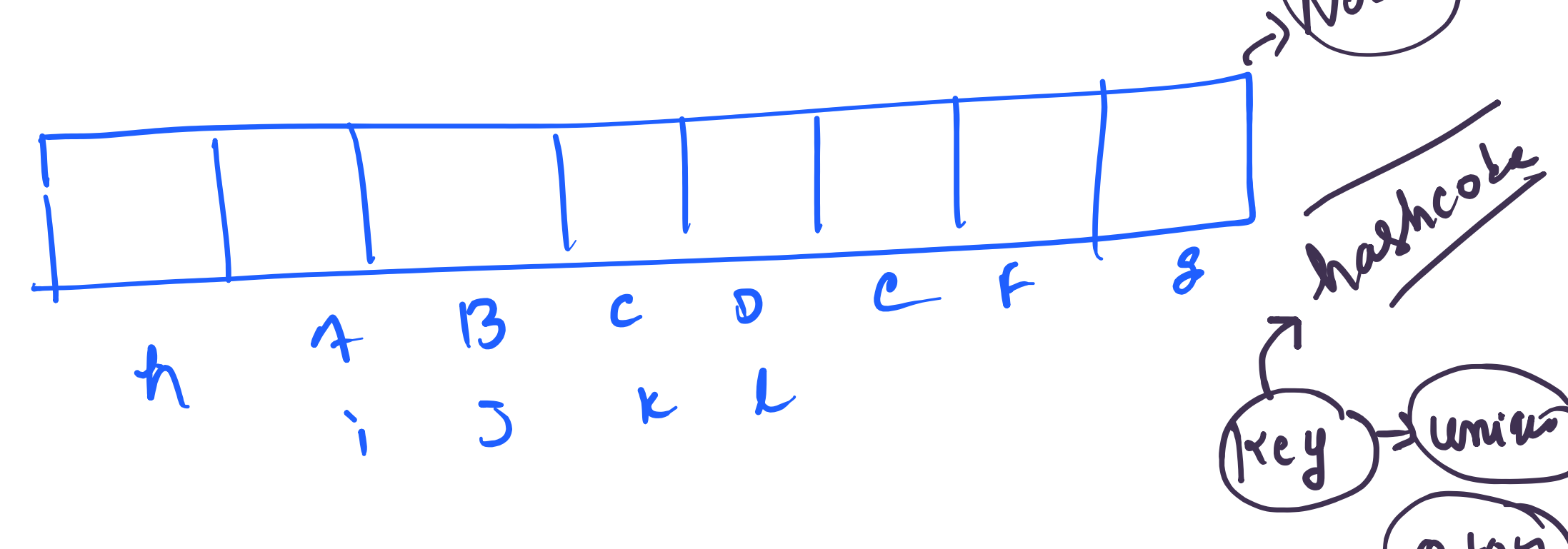
⑪ get, remove, contains



① $thf = 2.0$
 $loadf = 1$

$$m \div 4 = 17$$

① Re-hashing $\rightarrow$?



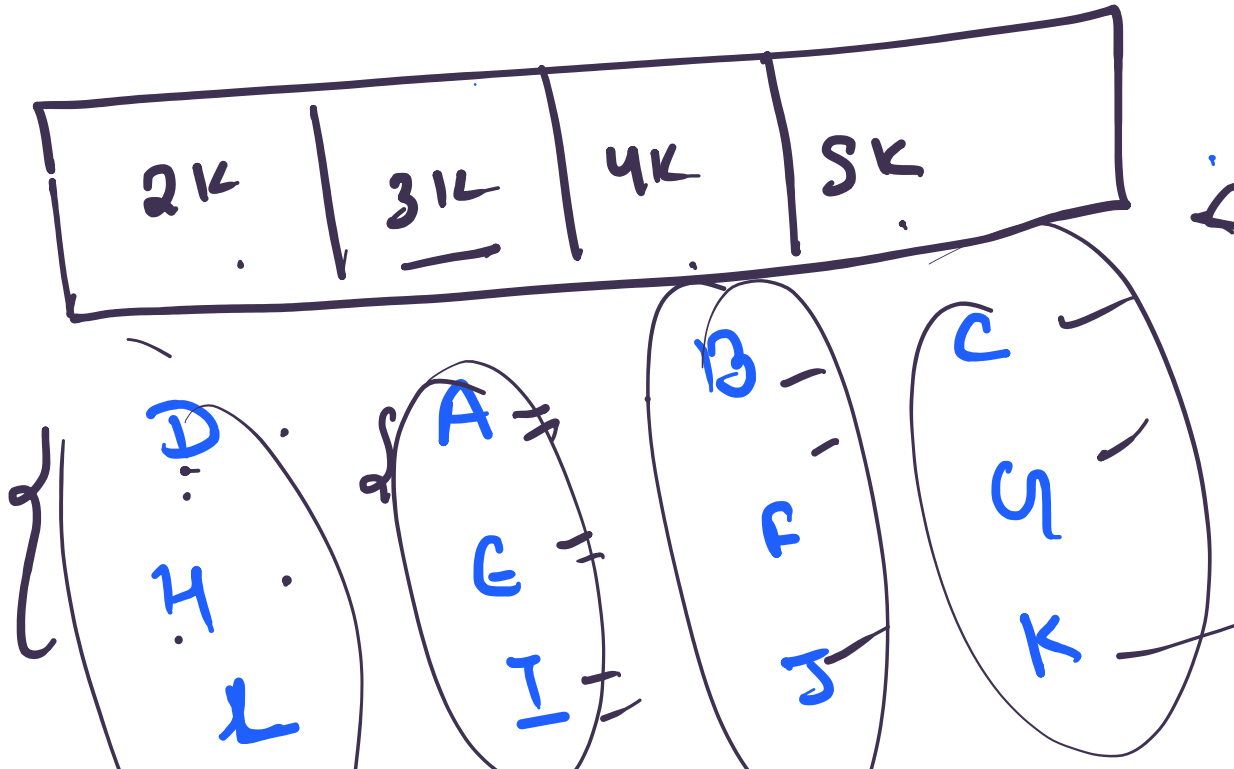
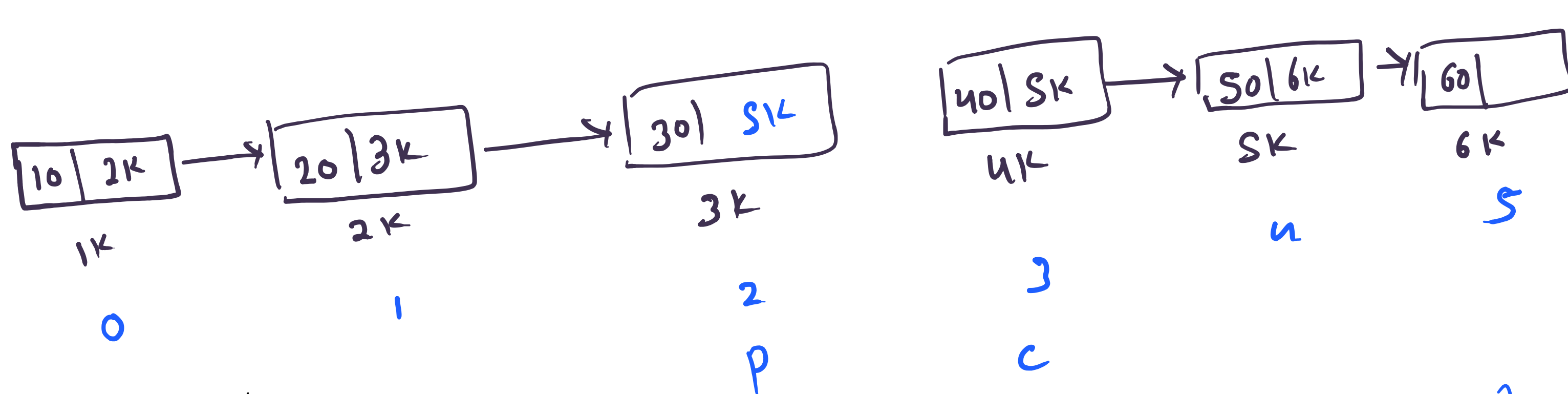
```

class Node {
    ← key
    ← value
    Node next
}

```

$$q_0 + ax + bx^2 + cx^3 + dx^4 \dots$$

Robinson

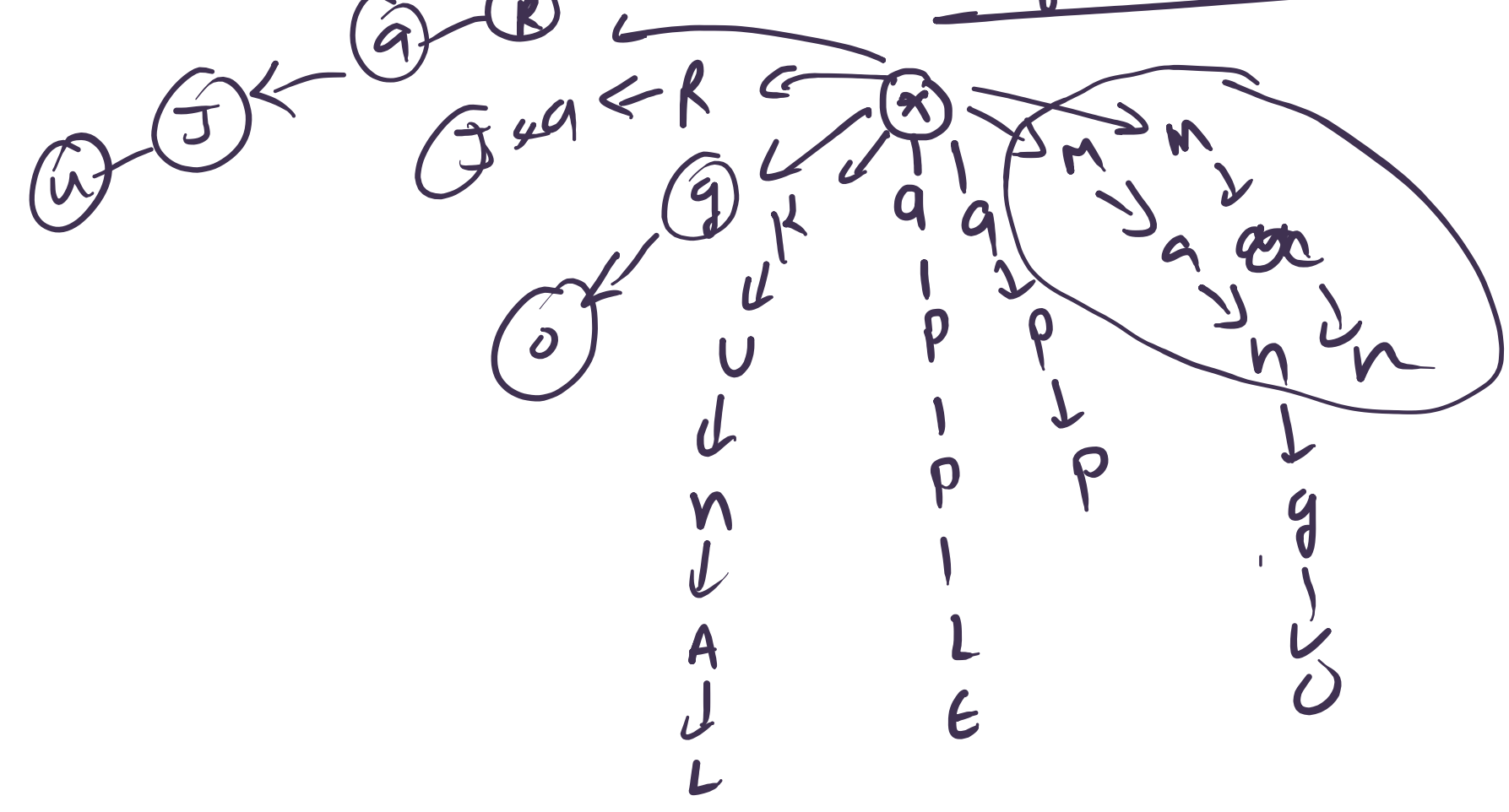


```
ArrayList<Node> new_list = new ArrayList<>();  
ArrayList<Node> old_list = list;  
list = new_list;
```

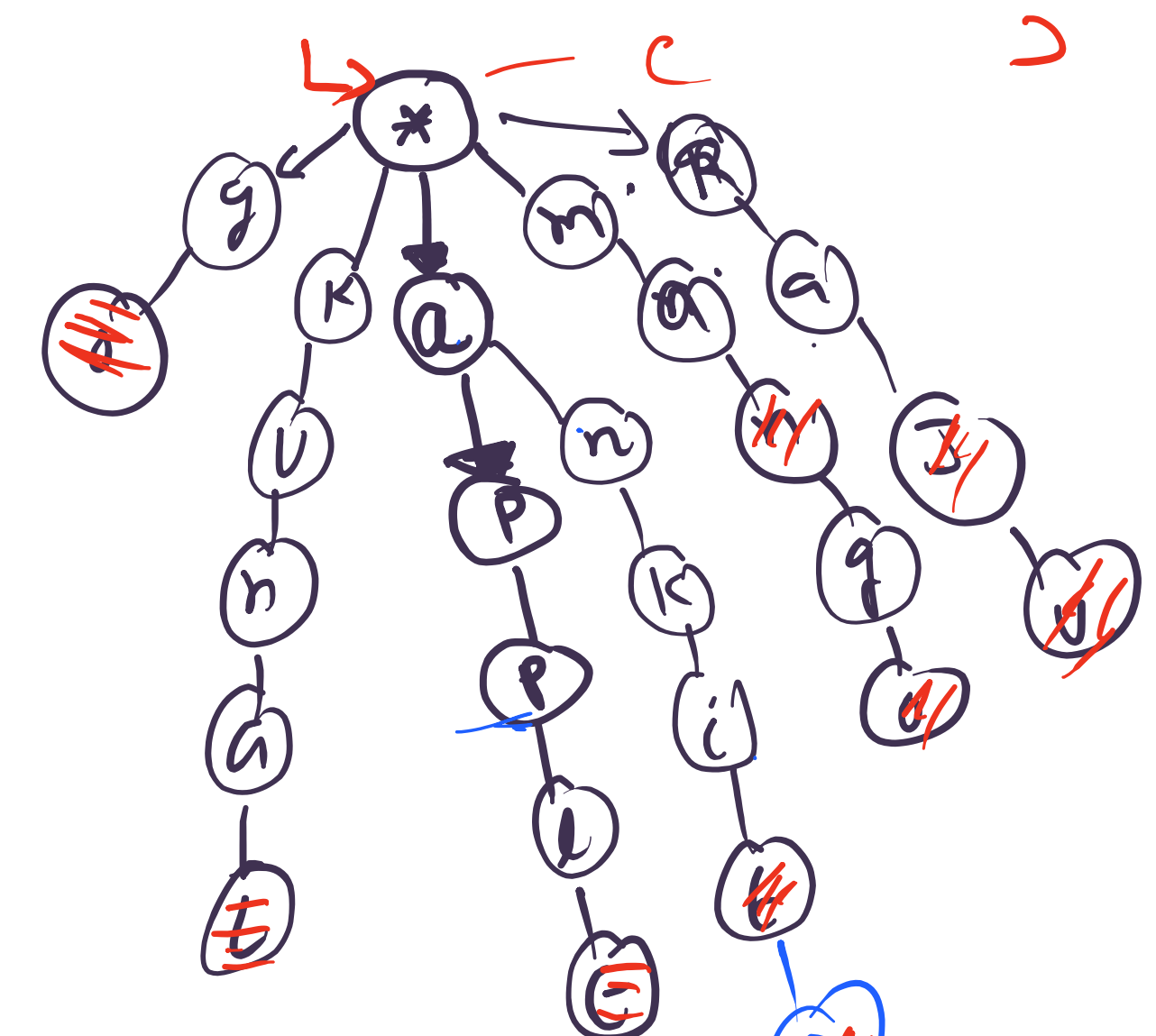
```
for (Node nn; oldList; ) {
    while (nn != null) {
        put (nn.key, nn.value);
        nn = nn.next;
    }
}
```

Muscle $\rightarrow$ um L

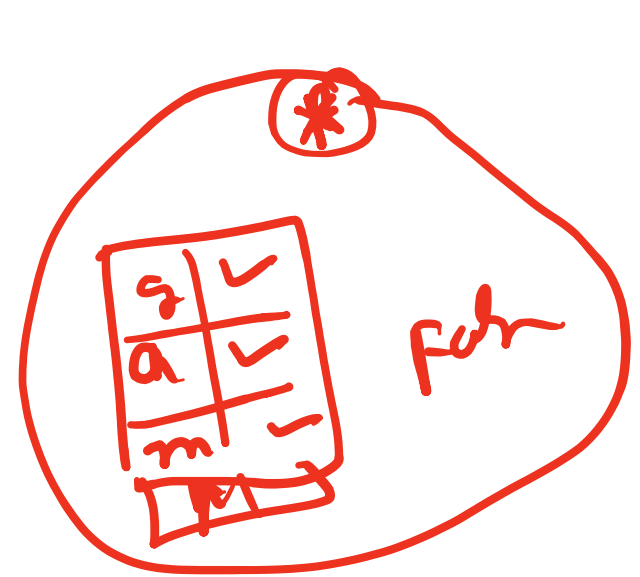
True $\Rightarrow$ Prefix True



① apple
 ② ankut
 ③ Kund
 ④ mango
 ⑤ go
 ⑥ man
 ⑦ Raj
 ⑧ Raja
 ankit ka



char ch
HashMap<ch, Node>
ch = 'c'

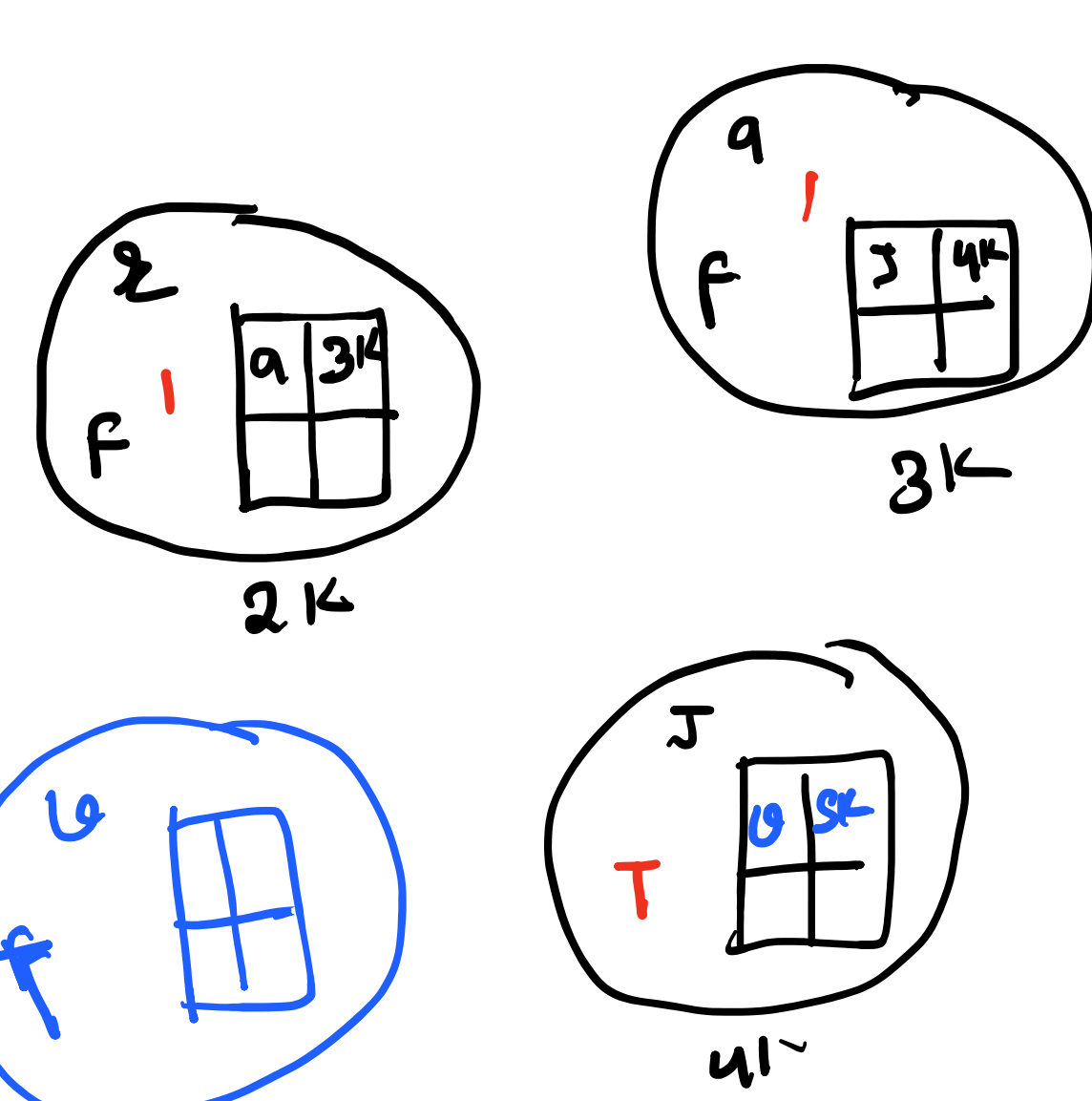


```

public void insert(String word) {
    Node curr=root;
    for (int i = 0; i < word.length(); i++) {
        char ch=word.charAt(i);
        if(curr.child.containsKey(ch)) {
            curr = curr.child.get(ch);
        }
        else {
            Node nn = new Node(ch);
            curr.child.put(ch, nn);
            curr=nn;
        }
    }
}

```

① sai ② saih

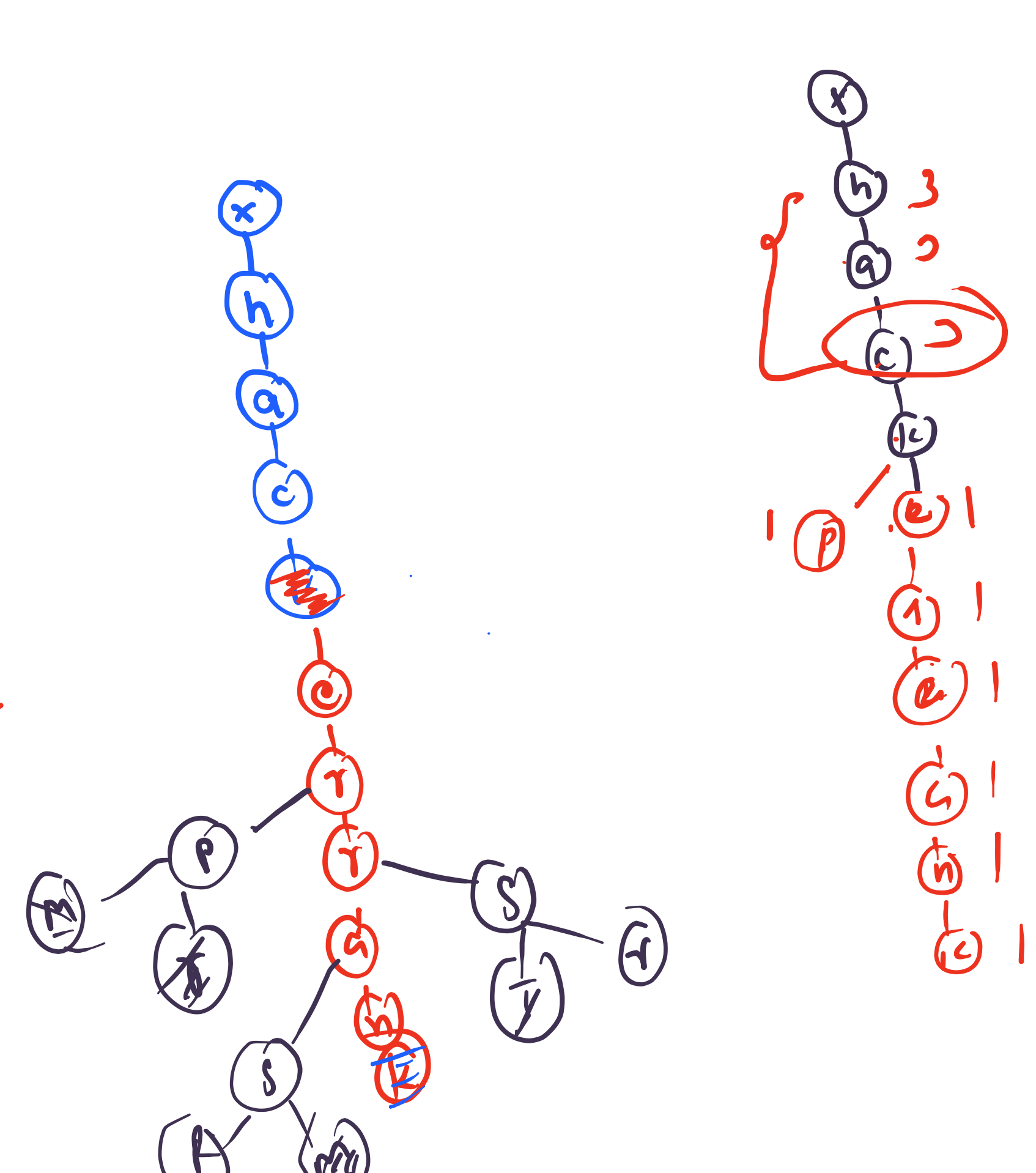


Sample Input

```
4
add hack
add hackerrank
find hac ✓
find hak
```

Sample Output

20



$2, 1, 1, 0, 1, 2, 5, 4, 0, 2, 8, 7, 9, 2, 6, 1, 9$
 $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16$

00111 2224567899

